Experimental data for C_2 (Carbon diatomic)

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Other names
Carbon (C2); Carbon dimer; Dicarbon;

INChI	INChIKey	SMILES	IUPAC name
InChI=1S/C2/c1-2	LBVWYGNGGJURHQ-UHFFFAOYSA-N	[C]#[C]	

State	Conformation
$^1\Sigma_g^+$	D*H

Enthalpy of formation (Hfg), Entropy, Integrated heat capacity (0 K to 298.15 K) (HH), Heat Capacity (Cp)

Property	Value	Uncertainty	units	Reference	Comment
Hfg(298.15K) $\Delta_f H°$	830.46	10.00	kJ mol^{-1}	Gurvich	
Hfg(0K) $\Delta_f H°$	822.39	10.00	kJ mol^{-1}	Gurvich	
Entropy (298.15K) $S°$	197.10		J K^{-1} mol^{-1}	Gurvich	
Integrated Heat Capacity (0 to 298.15K) $H°-H°(T)$	10.17		kJ mol^{-1}	Gurvich	
Heat Capacity (298.15K) $C_p°$	43.55		J K^{-1} mol^{-1}	Gurvich	

Information can also be found for this species in the NIST Chemistry Webbook

Vibrational levels (cm^{-1})

Mode Number	Symmetry	Frequency			Intensity			Comment	Description
		Fundamental(cm^{-1})	Harmonic(cm^{-1})	Reference	(km mol^{-1})	unc.	Reference		
1	Σ_g	1827	1855	2007Iri:389					

Detailed diatomic data

ω_e	$\omega_e x_e$	$\omega_e y_e$	B_e	α_e	ZPE	reference
1855.066	13.6007	-0.116	1.820053	0.0179143	923.9669	2007Iri:389

vibrational zero-point energy: 913.7 cm^{-1} (from fundamental vibrations)
Calculated vibrational frequencies for C_2 (Carbon diatomic).

Rotational Constants (cm^{-1})

See section I.F.4 to change rotational constant units

A	B	C	reference	comment
	1.82005		2007Iri:389	

Calculated rotational constants for C_2 (Carbon diatomic).

Product of moments of inertia $I_A I_B I_C$				
9.262164	amu Å^2		1.538043E-39	gm cm^2

Geometric Data

Point Group $D_{\infty h}$

Internal coordinates

distances (r) in Å, angles (a) in degrees, dihedrals (d) in degrees

Description	Value	unc.	Connectivity				Reference	Comment
			Atom 1	Atom 2	Atom 3	Atom 4		
rCC	1.243		1	2			1979HUB/HER	re

Cartesians

Atom	x (Å)	y (Å)	z (Å)
C1	0.0000	0.0000	0.0000
C2	0.0000	0.0000	1.2425

Atom - Atom Distances

Distances in Å

	C1	C2
C1		**1.2425**
C2	**1.2425**	

Calculated geometries for C_2 (Carbon diatomic).

Bond descriptions

Examples: C-C single bond, C=C, double bond, C#C triple bond, C:C aromatic bond

Bond Type	Count
C#C	1

Connectivity

Atom 1	Atom 2
C1	C2

Electronic energy levels (cm^{-1})

Energy (cm^{-1})	Degeneracy	reference	description
0	1	Gurvich	$^1\Sigma_g^+$
716.24	6	Gurvich	$^3\Pi_u$
6434.27	3	Gurvich	$^3\Sigma_g^-$
8391	2	Gurvich	$^1\Pi_u$

Ionization Energies (eV)

Ionization Energy	I.E. unc.	vertical I.E.	v.I.E. unc.	reference
11.400	0.300			webbook

Electron Affinity (eV)

Electron Affinity	unc.	reference
3.273	0.008	webbook

Dipole, Quadrupole and Polarizability

Electric dipole moment

State	Config	State description	Conf description	Exp. min.	Dipole (Debye)				Reference	comment	Point Group	Components	
					x	y	z	total				dipole	quadrupole
1	1	$^1\Sigma_g^+$	$D_{\infty h}$	True				0.000	NSRDS-NBS10		$D_{\infty h}$	0	1
2	1	$^3\Pi_u$	$D_{\infty h}$	True							$D_{\infty h}$	0	1
3	1	$^3\Sigma_g^-$	$D_{\infty h}$	True							$D_{\infty h}$	0	1
4	1	$^1\Pi_u$	$D_{\infty h}$	True							$D_{\infty h}$	0	1

Experimental dipole measurement abbreviations: MW microwave; DT Dielectric with Temperature variation; DR Indirect (usually an upper limit); MB Molecular beam

Calculated electric dipole moments for C_2 (Carbon diatomic).

Electric quadrupole moment

State	Config	State description	Conf description	Exp. min.	Quadrupole (D Å)			Reference	comment	Point Group	Components	
					xx	yy	zz				dipole	quadrupole
1	1	$^1\Sigma_g^+$	$D_{\infty h}$	True						$D_{\infty h}$	0	1
2	1	$^3\Pi_u$	$D_{\infty h}$	True						$D_{\infty h}$	0	1
3	1	$^3\Sigma_g^-$	$D_{\infty h}$	True						$D_{\infty h}$	0	1
4	1	$^1\Pi_u$	$D_{\infty h}$	True						$D_{\infty h}$	0	1

Calculated electric quadrupole moments for C_2 (Carbon diatomic).

References

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squib	reference	DOI
1979HUB/HER	Huber, K.P.; Herzberg, G., Molecular Spectra and Molecular Structure. IV. Constants of Diatomic Molecules, Van Nostrand Reinhold Co., 1979	10.1007/978-1-4757-0961-2
2007Iri:389	KK Irikura "Experimental Vibrational Zero-Point Energies: Diatomic Molecules" J. Phys. Chem. Ref. Data 36(2), 389, 2007	10.1063/1.2436891
Gurvich	Gurvich, L.V.; Veyts, I. V.; Alcock, C. B., Thermodynamic Properties of Individual Substances, Fouth Edition, Hemisphere Pub. Co., New York, 1989	
NSRDS-NBS10	R. D. Nelson Jr., D. R. Lide, A. A. Maryott "Selected Values of electric dipole moments for molecules in the gas phase" NSRDS-NBS10, 1967	10.6028/NBS.NSRDS.10
webbook	NIST Chemistry Webbook (http://webbook.nist.gov/chemistry)	10.18434/T4D303

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BSi (Boron silicide)	C2- (carbon diatomic anion)